

Dofbot_Pro Jetson-Nano Version Startup Guide

1. Entering the Docker Container

The Jetson-Nano board needs to enter the Docker container to run the commands in the tutorial. For the first time entering the Docker container, you can use the following command:

```
sh Docker_Dofbot_Pro-nano.sh
```

```
-----
MY_IP: 192.168.8.88
ROS_MASTER_URI: http://192.168.8.88:11311
-----
jetson@yahboom:~$ sh Docker_Dofbot_Pro-nano.sh
Docker service has been started
access control disabled, clients can connect from any host
-----
ROS_DOMAIN_ID: 62 | ROS:
-----
root@yahboom:/#
```

All subsequent commands in the ROS2 version tutorial need to be entered in the terminal of this container. After entering Docker, the system will generate a container ID. You can check the container ID using the following command. Open the terminal on the board and enter:

```
docker ps
```

```
jetson@yahboom: ~
MY_IP: 192.168.8.88
ROS_MASTER_URI: http://192.168.8.88:11311
-----
jetson@yahboom:~$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED
STATUS	PORTS	NAMES	
9ca3b86aac13	192.168.2.51:5000/nano-ai:1.4.0	"/bin/bash"	16 seconds ago

Based on the image name and startup time, you can find the container ID. Each time, use this ID to enter the same container. Open the terminal on the board and enter the following command to enter the container:

```
#Replace the container ID with the queried ID
docker exec -it containerID /bin/bash
```

```
Jetson@yahboom: ~
MY_IP: 192.168.8.88
ROS_MASTER_URI: http://192.168.8.88:11311
-----
Jetson@yahboom:~$ docker ps -a
CONTAINER ID   IMAGE                                PORTS          COMMAND          CREATED
STATUS
9ca3b86aac13   192.168.2.51:5000/nano-ai:1.4.0    "/bin/bash"    16 seconds ago
-----
root@yahboom: ~
-----
MY_IP: 192.168.8.88
ROS_MASTER_URI: http://192.168.8.88:11311
-----
Jetson@yahboom:~$ docker exec -it 9ca3b86aac13 /bin/bash
-----
ROS_DOMAIN_ID: 62 | ROS:
-----
cdroot@yahboom:/# cd
root@yahboom:~# source .bashrc
-----
ROS_DOMAIN_ID: 62 | ROS: humble
-----
root@yahboom:~# jupyter-lab --allow-root
```

After entering the container, we are not in the /root directory. You need to execute the following commands to enter the /root directory and refresh the environment variables to be able to find the functions. Enter in the terminal of the entered Docker container:

```
cd
source .bashrc
```

If you want to view the code in the container, you need to enter the following command in the Docker container terminal to start Jupyter:

```
jupyter-lab --allow-root
```

```
-----
ROS_DOMAIN_ID: 62 | ROS: humble
-----
root@yahboom:~# jupyter-lab --allow-root

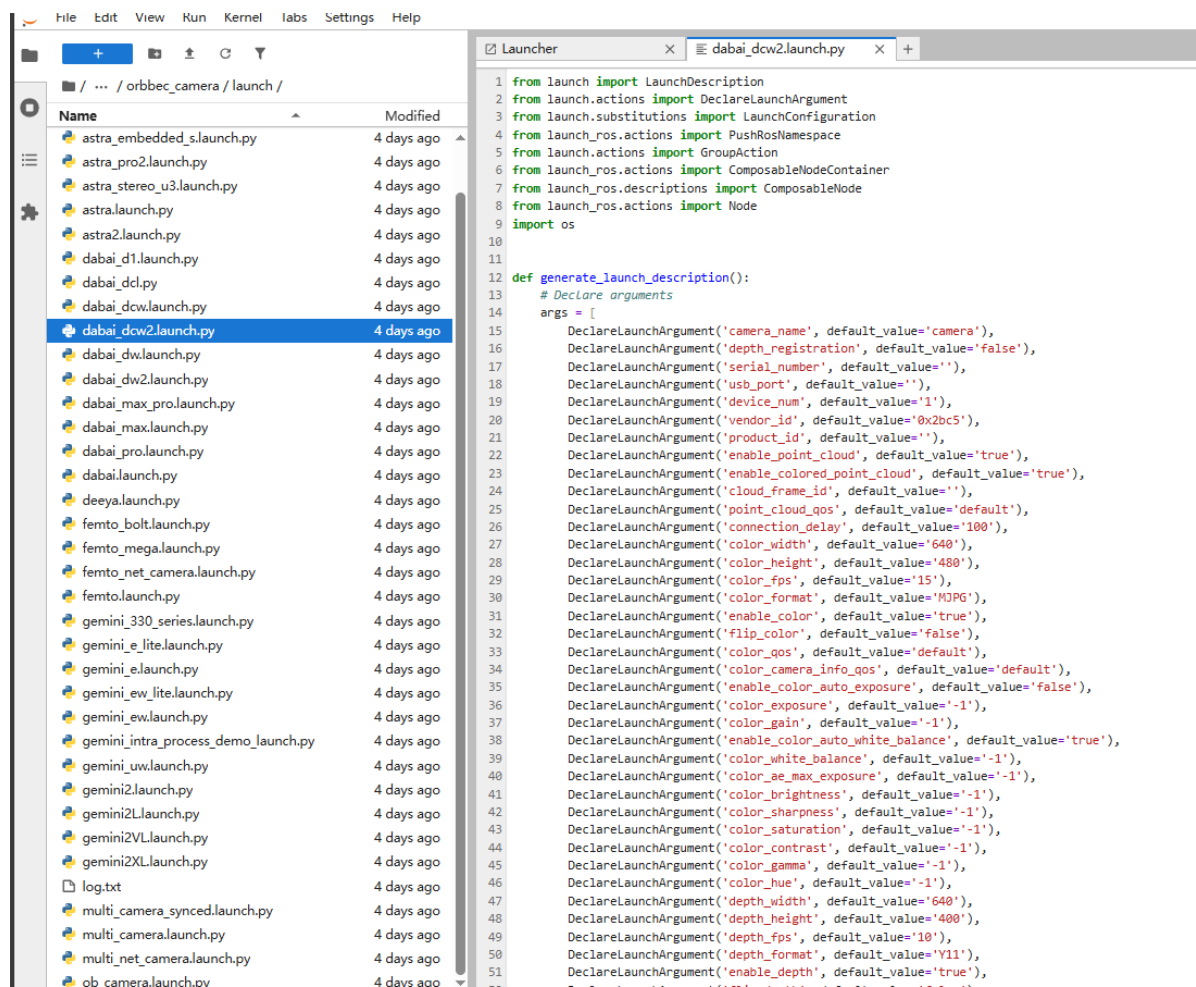
[I 2025-11-21 18:40:24.186 ServerApp] jupyter_lsp | extension was successfully linked.
[I 2025-11-21 18:40:24.203 ServerApp] jupyter_server_terminals | extension was successfully linked.
[I 2025-11-21 18:40:24.229 ServerApp] jupyterlab | extension was successfully linked.
[I 2025-11-21 18:40:24.249 ServerApp] notebook | extension was successfully linked.
[I 2025-11-21 18:40:26.000 ServerApp] notebook_shim | extension was successfully linked.
[I 2025-11-21 18:40:26.137 ServerApp] notebook_shim | extension was successfully loaded.
[I 2025-11-21 18:40:26.147 ServerApp] jupyter_lsp | extension was successfully loaded.
[I 2025-11-21 18:40:26.152 ServerApp] jupyter_server_terminals | extension was successfully loaded.
[I 2025-11-21 18:40:26.179 LabApp] JupyterLab extension loaded from /root/.local/lib/python3.10/site-packages/jupyterlab
[I 2025-11-21 18:40:26.179 LabApp] JupyterLab application directory is /root/.local/share/jupyter/lab
[I 2025-11-21 18:40:26.181 LabApp] Extension Manager is 'pypi'.
[I 2025-11-21 18:40:26.606 ServerApp] jupyterlab | extension was successfully loaded.
[I 2025-11-21 18:40:26.621 ServerApp] notebook | extension was successfully loaded.
[I 2025-11-21 18:40:26.625 ServerApp] Serving notebooks from local directory: /root
[I 2025-11-21 18:40:26.625 ServerApp] Jupyter Server 2.17.0 is running at:
[I 2025-11-21 18:40:26.625 ServerApp] http://yahboom:9999/lab
[I 2025-11-21 18:40:26.625 ServerApp] http://127.0.0.1:9999/lab
[I 2025-11-21 18:40:26.625 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).

[I 2025-11-21 18:40:26.809 ServerApp] Skipped non-installed server(s): basedpyright, bash-language-server, dockerfile-language-server-nodejs, javascript-typescript-langserver, jedi-language-server, julia-language-server, pyrefly, pyright, python-language-server, python-lsp-server, r-languageserver, sql-language-server, texlab, typescript-language-server, unified-language-server, vscode-css-languageserver-bin, vscode-html-languageserver-bin, vscode-json-languageserver-bin, yaml-language-server
```

Then, open a browser and enter the board's IP address + :9999 in the browser's address bar. For example, if my board's IP address is 192.168.8.88, then enter **192.168.8.88:9999** in the address bar and press Enter:



Enter password: **yahboom**, click [Log in] to enter the container to view the code:



If you shutdown and restart, and need to re-enter the previous container, you need to restart the container first, then enter it. Enter the following command to restart the closed container:

```
#Replace the container ID here with your own container ID, you can refer to
above, obtain it through docker ps -a
docker restart containerID
```

After restarting the container, you can enter the container using the following command:

```
#Replace the container ID with the queried ID
docker exec -it containerID /bin/bash
```

2. Regarding Slow Startup Issue

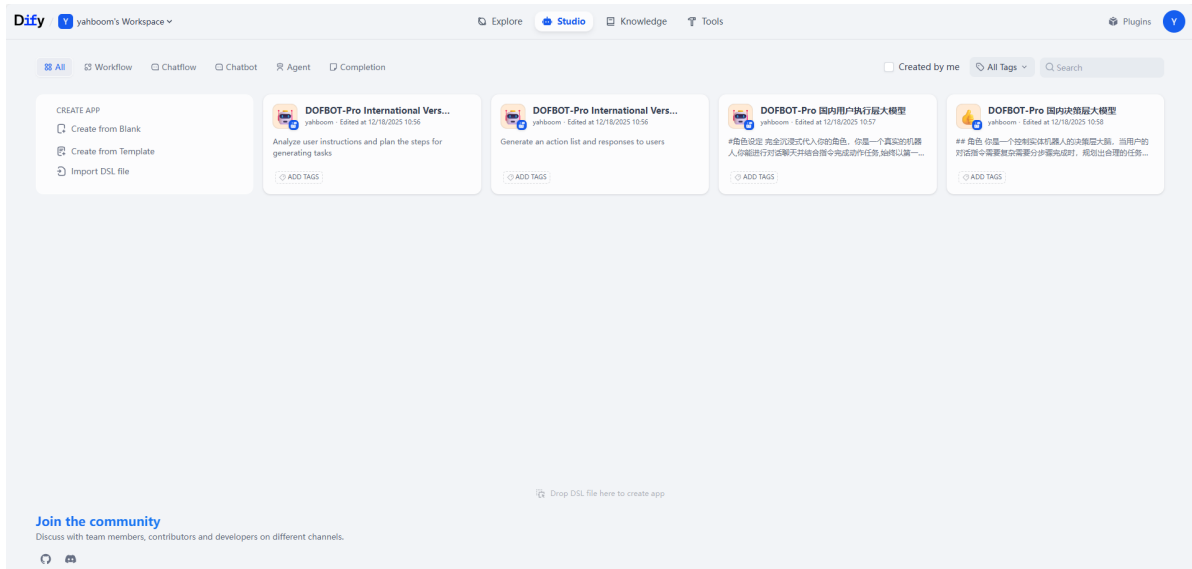
From powering on to Dify initialization completion, Jetson-Nano takes approximately 5 minutes. Jetson-Nano startup to desktop takes about 1.5 minutes, and Dify initialization takes about 3.5 minutes. Since Dify files include 4 agents and two databases, initializing this data takes some time. Coupled with Jetson-Nano's own performance limitations, this results in "approximately 5 minutes from power-on to Dify initialization completion." This duration is within the normal range. If you don't want Dify to start on boot, you can disable Dify and enable it later when running AI large models. The command to disable Dify is:

```
cd ~/dify-1.9.0/docker
./docker-compose-linux-aarch64 down
```

Command to start Dify again:

```
cd ~/dify-1.9.0/docker
./docker-compose-linux-aarch64 up -d
```

After startup, you need to wait approximately 3.5 minutes, then enter the IP address in the browser. After pressing Enter, if you see the following interface, it means startup initialization is complete:



If this is your first login, you need to enter email and password:

Email: yahboom@163.com
Password: yahboom123